Initial Scaling Analysis of Dynamic Speckle from a Rotating Diffuser and Possible Applications in Information Processing and Random Thermal Excitation

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Abstract

This document defines an initial analysis framework for intensity autocorrelation data obtained from coherent light scattered by a rotating multiple-scattering diffuser. The measured data consist of second-order intensity correlation curves at several scattering angles and rotation speeds. The first analysis goal is to test whether the temporal correlation curves collapse when time delay is replaced by diffuser angular displacement, $\Delta\theta = \omega\tau$. A stronger analysis incorporates the tangential displacement $R\omega\tau$ and a possible scattering-vector-dependent decorrelation length $\ell_c(q)$. The second part outlines possible applications in information transmission, random-feature optical computing, and spatially random thermal excitation for transport measurements. The document is intended both as a physics note and as a specification for analysis code.

Contents

1 Experimental variables and notation

A coherent optical beam illuminates a rotating diffuser at radial distance R from the axis of rotation. The diffuser rotates at angular speed ω. A detector measures scattered intensity at scattering angle ϕ, defined as the angle between the incident and detected wavevectors.

The magnitude of the scattering vector is

$$\boxed{q = \frac{4\pi}{\lambda} \sin\left(\frac{\phi}{2}\right)} \tag{1}$$

where λ is the optical wavelength in the surrounding medium. If the beam propagates in a material with refractive index n, replace λ by the wavelength in the medium, λ_0/n, or write

$$q = \frac{4\pi n}{\lambda_0} \sin\left(\frac{\phi}{2}\right). \tag{2}$$

The primary measured quantity is the intensity time series

$$I(t, q; \omega, R), \tag{3}$$

or, when the diffuser angle is measured,

$$I(\theta, q), \qquad \theta(t) = \theta_0 + \omega t \tag{4}$$

for uniform rotation.

The present data consist primarily of normalized second-order intensity autorrelation functions rather than complete intensity traces.

2 Definition of the intensity autocorrelation

For a detector at scattering vector q, define

$$g^{(2)}(\tau, q; \omega, R) = \frac{\langle I(t, q; \omega, R) I(t + \tau, q; \omega, R)\rangle_t}{\langle I(t, q; \omega, R)\rangle_t^2}. \tag{5}$$

For a stationary intensity process,

$$g^{(2)}(\tau, q; \omega, R) \to 1 \tag{6}$$

at delays much longer than the correlation time.

It is often useful to remove differences in measured speckle contrast by defining

$$\boxed{G^{(2)}(\tau, q; \omega, R) = \frac{g^{(2)}(\tau, q; \omega, R) - 1}{g^{(2)}(0, q; \omega, R) - 1}.} \tag{7}$$

Ideally,

$$G^{(2)}(0, q; \omega, R) = 1, \qquad G^{(2)}(\tau \to \infty, q; \omega, R) = 0. \tag{8}$$

The normalization in Eq. (7) removes the contrast factor from the comparison of decay shapes. The raw value $g^{(2)}(0) - 1$ should nevertheless be retained and analyzed separately because it contains information about detector area, polarization averaging, spatial coherence, and the number of unresolved speckle modes.

3 Candidate functional form

A useful empirical model is the stretched or compressed exponential

$$g^{(2)}(\tau, q; \omega, R) - 1 = \beta(q, \omega, R) \exp\left[-2\left|\frac{\tau}{\tau_c(q, \omega, R)}\right|^{\alpha}\right], \tag{9}$$

where

- $\beta = g^{(2)}(0) - 1$ is the measured correlation contrast,
- τ_c is a characteristic correlation time,
- α controls the decay shape.

The factor of 2 in Eq. (9) is conventional when the intensity correlation is related to the square of a field correlation. The analysis code should allow this factor to be removed or absorbed into the definition of τ_c.

The exponent should initially remain a fitted parameter. Common limiting cases include

$$\alpha = 1 \qquad \text{exponential decay,} \tag{10}$$

$$\alpha = 2 \qquad \text{Gaussian decay.} \tag{11}$$

The main scaling tests should not rely exclusively on fitted values of τ_c. Full-curve collapse tests use more information and can reveal changes in shape, shoulders, oscillations, or multiple timescales.

4 First prediction: collapse versus $\omega\tau$

4.1 Angular displacement as the governing variable

During a delay τ, a uniformly rotating diffuser advances through angle

$$\Delta\theta = \omega\tau. \tag{12}$$

If rotation speed changes only the rate at which a fixed sequence of diffuser configurations is traversed, then

$$I(t, q; \omega, R) = I\left(\theta_0 + \omega t, q; R\right). \tag{13}$$

The normalized correlation should then have the form

$$\boxed{G^{(2)}(\tau, q; \omega, R) = F_{q,R}(\omega\tau).} \tag{14}$$

At fixed q and fixed R, curves measured at different ω should collapse when plotted against

$$u_\theta = \omega\tau. \tag{15}$$

The correlation time must consequently scale as

$$\boxed{\tau_c(q, \omega, R) = \frac{\Delta\theta_c(q, R)}{\omega},} \tag{16}$$

where $\Delta\theta_c(q, R)$ is the characteristic angular displacement required to decorrelate the detected intensity.

Thus, at fixed q and R,

$$\tau_c \propto \omega^{-1}. \tag{17}$$

A plot of τ_c versus $1/\omega$ should be approximately linear and, within uncertainty, should pass through the origin unless there is an additional speed-independent decorrelation process.

4.2 What successful collapse means

Successful collapse versus $\omega\tau$ supports the following interpretation:

> The normalized speckle dynamics are governed primarily by diffuser angular displacement. Rotation speed changes the temporal playback rate but does not substantially change the underlying normalized stochastic process.

This is stronger than finding only $\tau_c \propto 1/\omega$, because collapse tests the entire shape of every correlation curve.

It does not prove that intensity traces at different q values are identical or mutually predictable. It establishes a scaling law for their normalized second-order statistics.

4.3 Failure of the first collapse

Systematic failure of collapse may indicate

- speed-dependent vibration or shaft wobble,
- nonuniform angular velocity,
- mechanical resonances,
- detector bandwidth limitations,
- time-bin or correlator artifacts,
- changing beam position on the diffuser,
- multiple decorrelation processes,
- diffuser heating or deformation,
- insufficient statistical averaging.

Failure is therefore diagnostically useful.

5 Second prediction: collapse versus tangential displacement

The material under the illuminated spot moves tangentially through distance

$$\Delta x = R\Delta\theta = R\omega\tau. \tag{18}$$

If decorrelation is governed by physical translation of the scattering structure through the illuminated region, the stronger scaling hypothesis is

$$\boxed{G^{(2)}(\tau, q; \omega, R) = F_q(R\omega\tau).} \tag{19}$$

Introduce a characteristic physical decorrelation length $\ell_c(q)$:

$$\boxed{G^{(2)}(\tau, q; \omega, R) = F\left(\frac{R\omega\tau}{\ell_c(q)}\right).} \tag{20}$$

The corresponding correlation time is

$$\boxed{\tau_c(q, \omega, R) = \frac{\ell_c(q)}{R\omega}.} \tag{21}$$

Therefore,

$$R\omega\tau_c = \ell_c(q). \tag{22}$$

Equation (22) provides an operational way to extract a physical displacement scale from each measured correlation curve.

At fixed q, the following predictions can be tested:

$$\tau_c \propto \omega^{-1}, \tag{23}$$

$$\tau_c \propto R^{-1}, \tag{24}$$

$$R\omega\tau_c \approx \text{constant}. \tag{25}$$

6 Possible dependence on scattering vector

The most conservative analysis does not assume a particular q dependence. Instead, determine

$$\ell_c(q) = R\omega\tau_c(q, \omega, R) \tag{26}$$

from the data and then examine the functional dependence of ℓ_c on q.

The general collapse variable is

$$\boxed{u = \frac{R\omega\tau}{\ell_c(q)}.} \tag{27}$$

Candidate models include the following.

6.1 No measurable q dependence

If

$$\ell_c(q) \approx \ell_0, \tag{28}$$

then

$$G^{(2)} = F\left(\frac{R\omega\tau}{\ell_0}\right). \tag{29}$$

All scattering channels have the same physical decorrelation displacement, although their actual intensity traces may still be different.

6.2 Inverse-q displacement scale

A phase-sensitivity argument may suggest

$$\ell_c(q) \propto \frac{1}{q}. \tag{30}$$

Then the dimensionless scaling variable is

$$u_q = qR\omega\tau, \tag{31}$$

and

$$\tau_c \propto \frac{1}{qR\omega}. \tag{32}$$

This model should be treated as a hypothesis, not assumed in advance, especially for a multiple-scattering diffuser whose dynamics may not reduce to a single-scattering phase factor.

6.3 General power law

A more flexible model is

$$\ell_c(q) = \ell_0 \left(\frac{q}{q_0}\right)^{-\nu}, \tag{33}$$

which gives

$$\tau_c(q, \omega, R) = \frac{\ell_0}{R\omega} \left(\frac{q}{q_0}\right)^{-\nu}. \tag{34}$$

The master variable becomes

$$u = \frac{R\omega\tau}{\ell_0} \left(\frac{q}{q_0}\right)^{\nu}. \tag{35}$$

The reference value q_0 keeps the expression dimensionally consistent. The exponent ν should be estimated from the data.

7 Revolution-synchronized recurrence

The first-collapse prediction concerns normalized statistics. A distinct question is whether the detailed intensity pattern repeats after one revolution.

The revolution period is

$$T_{\mathrm{rev}} = \frac{2\pi}{\omega}. \tag{36}$$

Perfect recurrence would imply

$$I(t + T_{\mathrm{rev}}, q) = I(t, q), \tag{37}$$

or equivalently,

$$I(\theta + 2\pi, q) = I(\theta, q). \tag{38}$$

Experimentally one expects approximate recurrence,

$$I(\theta + 2\pi, q) \approx I(\theta, q), \tag{39}$$

limited by photon statistics, laser noise, diffuser runout, mechanical drift, and imperfect angle registration.

A useful revolution-to-revolution recurrence coefficient is

$$\rho_{\mathrm{rev}}(q) = \frac{\langle \delta I_r(\theta, q) \delta I_{r+1}(\theta, q) \rangle_\theta}{\sqrt{\langle \delta I_r^2 \rangle_\theta \left\langle \delta I_{r+1}^2 \right\rangle_\theta}}, \tag{40}$$

where r labels revolutions.

Collapse and recurrence answer different questions:

- Collapse tests whether the normalized statistics scale with angular or tangential displacement.
- Recurrence tests whether the detailed diffuser-indexed intensity pattern can be replayed and calibrated.

A system may show good collapse but poor recurrence. It would then provide predictable stochastic statistics but not a stable, deterministic sequence of optical features.

8 Recommended initial analysis workflow

The following sequence is intended as a direct guide for analysis code.

8.1 Input data

For each experimental run, store

- delay values τ_i,
- measured $g^{(2)}(\tau_i)$,
- uncertainty or number of averages, when available,

- scattering angle ϕ,
- wavelength λ,
- calculated q from Eq. (1),
- angular speed ω,
- beam radius R on the diffuser,
- detector aperture and optical configuration,
- acquisition duration,
- count rate or mean intensity,
- rotation direction.

Use SI units internally:

$$q\,[\mathrm{m}^{-1}], \quad R\,[\mathrm{m}], \quad \omega\,[\mathrm{rad\,s}^{-1}], \quad \tau\,[\mathrm{s}]. \tag{41}$$

8.2 Preprocessing

For each curve:

1. Estimate the long-delay baseline.
2. Correct the baseline to unity if justified.
3. Calculate $\beta = g^{(2)}(0) - 1$.
4. Calculate $G^{(2)}$ using Eq. (7).
5. Preserve both raw and normalized curves.
6. Flag curves that do not reach a stable long-delay baseline.

Do not force the final measured point to equal one. Estimate the baseline from a delay region that appears statistically stationary.

8.3 Individual-curve fits

Fit each curve to Eq. (9), initially allowing

$$\beta, \quad \tau_c, \quad \alpha, \quad \text{and baseline} \tag{42}$$

to vary.
Also compare restricted models with $\alpha = 1$ and $\alpha = 2$.
Report

- fitted parameters,
- parameter uncertainties,

- residuals,
- reduced chi-squared or another goodness-of-fit measure,
- information criteria for competing models,
- the delay range included in the fit.

8.4 First-collapse plot

For each fixed q and R, plot

$$G^{(2)} \quad \text{versus} \quad \omega\tau. \tag{43}$$

The plotting code should show all rotation speeds on the same axes.

Quantify collapse error by interpolating the curves onto a shared dimensionless grid and calculating, for example,

$$\chi^2_{\text{collapse}} = \frac{1}{N_{\text{tot}}} \sum_{j,i} \frac{\left[G_j^{(2)}(u_i) - \overline{G^{(2)}}(u_i)\right]^2}{\sigma^2_{j,i}}, \tag{44}$$

where j labels runs and $\overline{G^{(2)}}$ is the estimated master curve.

If reliable pointwise uncertainties are unavailable, use an unweighted RMS collapse error and estimate uncertainty by bootstrap resampling.

8.5 Correlation-time scaling

At fixed q and R, fit

$$\tau_c = \frac{A}{\omega} + B. \tag{45}$$

The ideal angular-displacement model predicts

$$B = 0, \qquad A = \Delta\theta_c. \tag{46}$$

A statistically significant nonzero B may indicate an additional timescale or a systematic analysis offset.

8.6 Tangential-displacement collapse

When multiple R values are available, plot

$$G^{(2)} \quad \text{versus} \quad R\omega\tau. \tag{47}$$

Calculate

$$\ell_c(q) = R\omega\tau_c \tag{48}$$

for each run. At fixed q, test whether the extracted values agree within uncertainty.

8.7 q-dependent collapse

Plot

$$\ell_c(q) = R\omega\tau_c \tag{49}$$

against q. Compare at least

$$\ell_c(q) = \ell_0, \tag{50}$$

$$\ell_c(q) = \frac{A}{q}, \tag{51}$$

$$\ell_c(q) = \ell_0 \left(\frac{q}{q_0}\right)^{-\nu}. \tag{52}$$

Only after estimating $\ell_c(q)$ should all data be plotted against

$$u = \frac{R\omega\tau}{\ell_c(q)}. \tag{53}$$

The strongest collapse claim would be

$$\boxed{G^{(2)}(\tau, q; \omega, R) = F\left(\frac{R\omega\tau}{\ell_c(q)}\right)} \tag{54}$$

over the measured ranges of q, ω, and R.

8.8 Useful plots

The initial analysis package should generate:

1. Raw $g^{(2)}(\tau)$ curves grouped by q.
2. Normalized $G^{(2)}(\tau)$ curves.
3. Fits and fit residuals.
4. τ_c versus $1/\omega$ at fixed q and R.
5. $G^{(2)}$ versus $\omega\tau$.
6. $G^{(2)}$ versus $R\omega\tau$, when R varies.
7. $\ell_c(q) = R\omega\tau_c$ versus q.
8. Global collapse versus $R\omega\tau/\ell_c(q)$.
9. α versus q, ω, and R.
10. β versus q, ω, and R.
11. Collapse residuals versus scaled delay.

9 Autocorrelation, spectra, and information

Define the centered intensity fluctuation

$$\delta I(t,q) = I(t,q) - \langle I(t,q)\rangle . \tag{55}$$

Its autocovariance is

$$C_{qq}(\tau) = \langle \delta I(t,q)\delta I(t+\tau,q)\rangle . \tag{56}$$

It is related to $g^{(2)}$ by

$$C_{qq}(\tau) = \langle I(q)\rangle^2 \left[g^{(2)}(\tau,q) - 1\right] . \tag{57}$$

By the Wiener–Khinchin theorem, the Fourier transform of the autocovariance is the intensity-fluctuation power spectral density:

$$S_{qq}(f) = \int_{-\infty}^{\infty} C_{qq}(\tau)e^{-\mathrm{i}2\pi f\tau}\mathrm{d}\tau. \tag{58}$$

Thus $g^{(2)}(\tau,q)$ determines the spectral power at that detector, apart from normalization and the DC contribution.

However, it does not determine the complex Fourier phase of the original intensity trace. Many different intensity realizations can have the same autocorrelation and power spectrum.

Consequently, universal autocorrelation scaling can coexist with a large information-bearing space:

> The family of signals may have a simple low-dimensional statistical description while the individual intensity realizations remain complex, distinct, and potentially high dimensional.

10 Multiple detectors and cross-correlation

For detectors at q_1 and q_2, define

$$I_1(t) = I(t,q_1), \tag{59}$$
$$I_2(t) = I(t,q_2). \tag{60}$$

The cross-covariance is

$$C_{12}(\tau) = \langle \delta I_1(t)\delta I_2(t+\tau)\rangle . \tag{61}$$

The normalized cross-correlation is

$$\rho_{12}(\tau) = \frac{C_{12}(\tau)}{\sqrt{C_{11}(0)C_{22}(0)}}. \tag{62}$$

The Fourier transform gives the cross-spectrum

$$S_{12}(f) = \int_{-\infty}^{\infty} C_{12}(\tau)e^{-\mathrm{i}2\pi f\tau}\mathrm{d}\tau. \tag{63}$$

Unlike an autospectrum, S_{12} is generally complex. Its phase describes the relative spectral phase between the two detector signals.

A useful normalized quantity is the magnitude-squared coherence,

$$\gamma_{12}^2(f) = \frac{|S_{12}(f)|^2}{S_{11}(f)S_{22}(f)}, \qquad 0 \le \gamma_{12}^2 \le 1. \tag{64}$$

Low coherence indicates that the detectors carry largely distinct second-order information at that frequency. High coherence indicates strong redundancy or a common physical component.

Two autocorrelation functions may collapse to the same master form after rescaling even while the detector signals themselves are not identical. Collapse concerns the form of the statistics. Cross-correlation concerns the shared information between channels.

11 Lag direction and causal interpretation

The cross-correlation in Eq. (61) is directional in its lag convention:

$$C_{12}(\tau) = \langle \delta I_1(t) \delta I_2(t+\tau) \rangle . \tag{65}$$

A peak at positive τ means that variations at detector 1 tend to precede similar variations at detector 2 by τ. A peak at negative τ means the opposite ordering.

For example, if

$$I_2(t) \approx a I_1(t - \tau_d) + \eta(t), \tag{66}$$

then the cross-correlation will generally peak near $\tau = \tau_d$, subject to the sign convention.

This may reveal

- propagation delays,
- speckle translation across detector positions,
- ordered traversal of scattering channels,
- thermal transport delays,
- detector or electronics delays,
- common-input responses with different transfer functions.

A nonzero-lag peak does not by itself prove physical causality. Common drivers, filtering, periodicity, or geometric motion can produce similar signatures. A causal claim requires a controlled perturbation, known timing, or an explicit dynamical model.

With a driven system, it is useful to correlate each detector with the known drive $u(t)$:

$$C_{uI_m}(\tau) = \langle \delta u(t) \delta I_m(t+\tau) \rangle . \tag{67}$$

The known drive supplies a causal time origin. The sign and shape of C_{uI_m} can then be interpreted as an input–output response more reliably than the detector–detector correlation alone.

12 Part II: Motivation and possible applications

13 Random optical feature generation

Let an optical input be represented by a vector $\boldsymbol{x}$. It may encode

- amplitudes in several spatial modes,
- an image or spatial mask,
- phases imposed by an SLM,
- several wavelength channels,
- polarization components,
- several temporal input channels.

For a diffuser angle θ and output channel q_m, a schematic field model is

$$E(\theta, q_m; \boldsymbol{x}) = \sum_j T_{mj}(\theta) x_j, \tag{68}$$

where $T_{mj}(\theta)$ is an angle-dependent optical transfer coefficient. The measured intensity is

$$I(\theta, q_m; \boldsymbol{x}) = \left| \sum_j T_{mj}(\theta) x_j \right|^2. \tag{69}$$

Linear optical propagation followed by square-law detection produces a nonlinear feature map of the input.

A finite feature vector may be assembled from several detector channels and several diffuser angles:

$$\boldsymbol{f}(\boldsymbol{x}) = \{I(\theta_n, q_m; \boldsymbol{x})\}_{n,m}. \tag{70}$$

A trained linear readout can then form

$$\widehat{\boldsymbol{y}} = W \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{b}. \tag{71}$$

The diffuser performs complicated optical mixing. Only the electronic readout needs to be trained.

13.1 Why collapse matters for computing

If

$$G^{(2)}(\tau, q; \omega, R) = F\left(\frac{R\omega\tau}{\ell_c(q)}\right), \tag{72}$$

then the normalized feature dynamics are governed by a predictable scaling law. The feature-refresh time is approximately

$$\tau_c(q) = \frac{\ell_c(q)}{R\omega}. \tag{73}$$

The approximate feature-refresh rate is therefore

$$\mathcal{R}_f(q) \sim \frac{1}{\tau_c(q)} = \frac{R\omega}{\ell_c(q)}. \tag{74}$$

Changing ω changes the rate at which the diffuser configurations are traversed. If recurrence is strong, the device resembles a cyclic bank of reproducible random transformations indexed by diffuser angle.

Collapse does not prove high computational dimension. That requires measurements across multiple angles, detectors, and inputs, followed by covariance, singular-value, mutual-information, or task-performance analysis.

14 Possible application to information transmission

A controlled diffuser can produce a reproducible, complicated carrier indexed by θ and q. Information could be encoded in the incident field $\boldsymbol{x}$ and recovered from one or more scattered intensity streams.

Possible architectures include:

1. **Calibrated random projection.** The receiver learns the map between input symbols and output intensity features.
2. **Spread-spectrum-like encoding.** An input symbol is distributed over many angular positions, detector channels, or spectral components.
3. **Challenge–response transmission.** The diffuser configuration acts as a physical key or challenge variable.
4. **Low-detector-count reception.** One or a few photon-counting detectors acquire sequential angular features rather than a full camera image.
5. **Joint spatial and temporal multiplexing.** Several q channels are detected in parallel while rotation supplies sequential features.

The information rate cannot be inferred from τ_c alone. It also depends on

- detected photon rate,
- shot noise and background,
- recurrence fidelity,
- cross-channel redundancy,
- input alphabet,
- calibration drift,
- detector bandwidth,
- the conditioning of the input-to-feature map.

A first communication experiment could compare classification error for several imposed input patterns as the number of detector channels and angular samples increases.

15 Fiber-coupler architectures

A nominal 2×2 fiber coupler has four physical ports. Depending on how it is used, it can split, combine, or interfere two optical fields.

Several arrangements may be useful.

15.1 Signal and reference splitting

One branch can monitor incident laser power while another enters the diffuser experiment. Correlation with the reference detector can remove or identify common laser fluctuations.

Let $I_{\mathrm{ref}}(t)$ be the reference signal. Then

$$C_{\mathrm{ref},m}(\tau) = \langle \delta I_{\mathrm{ref}}(t) \delta I_m(t+\tau) \rangle . \tag{75}$$

This distinguishes input-source noise from diffuser-generated dynamics.

15.2 Two scattered channels

Fibers positioned at q_1 and q_2 can feed two detectors directly or be routed through a coupler. Direct detection preserves separate intensity streams. A coupler can instead form optical sums and differences before detection.

15.3 Interferometric combination

If coherent fields E_1 and E_2 enter the two input ports of an ideal balanced coupler, the output intensities contain interference terms:

$$I_+ \propto |E_1 + E_2|^2, \tag{76}$$

$$I_- \propto |E_1 - E_2|^2, \tag{77}$$

up to coupler phase conventions.

Their difference isolates a field-interference quadrature:

$$I_+ - I_- \propto 4 \,\mathrm{Re}\,(E_1 E_2^*) . \tag{78}$$

With an imposed phase shift, the complementary quadrature can be measured. Such a configuration can recover field-sensitive information that is absent from independent intensity measurements, provided coherence, polarization, and path stability are maintained.

A simple intensity cross-correlator and an optical interferometer are not equivalent. The first measures statistical relations between detected intensities. The second mixes optical fields before square-law detection.

15.4 Balanced detection

Two output ports of a balanced coupler can be sent to a balanced detector. This may suppress common-mode intensity noise and enhance a weak modulated or interferometric component.

For photon-counting operation, detector dead time, afterpulsing, timing jitter, and accidental coincidences must be included in the analysis.

16 Randomly driven thermal fields

A dynamic speckle field can also serve as a spatially and temporally random heating pattern. Let the absorbed optical power density be

$$P(\boldsymbol{r}, t) = A(\boldsymbol{r}) I_{\text{pump}}(\boldsymbol{r}, t), \tag{79}$$

where $A(\boldsymbol{r})$ represents optical absorption.

For small perturbations, the temperature response can be written as a linear convolution,

$$\Delta T(\boldsymbol{r}, t) = \int_{-\infty}^{t} \mathrm{d}t' \int \mathrm{d}^d r' \, \chi_T(\boldsymbol{r} - \boldsymbol{r}', t - t') P(\boldsymbol{r}', t'), \tag{80}$$

where χ_T is a causal thermal response function.

In Fourier space,

$$\Delta T(\boldsymbol{k}, \Omega) = \chi_T(\boldsymbol{k}, \Omega) P(\boldsymbol{k}, \Omega). \tag{81}$$

For ordinary Fourier diffusion in a homogeneous material,

$$\left(-\mathrm{i}\Omega C + \kappa k^2\right) \Delta T(\boldsymbol{k}, \Omega) = P(\boldsymbol{k}, \Omega), \tag{82}$$

so

$$\chi_T(\boldsymbol{k}, \Omega) = \frac{1}{\kappa k^2 - \mathrm{i}\Omega C}, \tag{83}$$

where C is volumetric heat capacity and κ is thermal conductivity.

The characteristic diffusive decay time at spatial frequency k is

$$\tau_{\text{th}}(k) = \frac{C}{\kappa k^2} = \frac{1}{D k^2}, \tag{84}$$

where $D = \kappa / C$ is thermal diffusivity.

A random heating field excites many $\boldsymbol{k}$ and Ω components simultaneously. If the incident random field is measured or reproducible, cross-correlation between the drive and the response can estimate the thermal transfer function over a range of spatial and temporal frequencies.

17 Random transient-grating transport measurements

A conventional transient-grating experiment uses a spatially periodic excitation with a selected grating wavevector. A random-transient-grating approach would instead use a known or measured random spatial pattern that contains a broad distribution of wavevectors.

Let

$$P(\boldsymbol{r}, t) \tag{85}$$

be the random optical heating field and let

$$Y_m(t) \tag{86}$$

be a probe observable measured at detector m. The observable might be

- diffracted probe intensity,

- transient reflectivity,
- transient transmission,
- optical phase,
- surface displacement,
- x-ray diffuse or Bragg scattering,
- acoustic response,
- electrical response coupled to temperature.

For a linear system,

$$Y_m(t) = \int_{-\infty}^{t} h_m(t-t')u(t')\mathrm{d}t' + \eta_m(t), \tag{87}$$

where $u(t)$ is a measured drive channel and h_m is the causal impulse response.

The input–output cross-spectrum is

$$S_{uY_m}(\Omega) = H_m(\Omega)S_{uu}(\Omega) \tag{88}$$

when the additive noise is uncorrelated with the drive. Therefore,

$$\boxed{H_m(\Omega) = \frac{S_{uY_m}(\Omega)}{S_{uu}(\Omega)}} \tag{89}$$

wherever the drive spectrum has adequate power and coherence.

This is a direct motivation for recording a reference copy of the random drive. A four-port fiber-coupler arrangement could send one portion of the pump-derived signal to a reference detector while another portion drives the sample.

17.1 Possible transport regimes

A broadband random thermal drive could probe:

- ordinary diffusive thermal transport,
- multiple thermal diffusion lengths in layered materials,
- anisotropic in-plane transport,
- interface thermal conductance,
- coupled carrier and phonon transport,
- thermoelastic acoustic generation,
- hydrodynamic or quasiballistic departures from Fourier diffusion,
- spatially heterogeneous transport,
- nonlinear response as drive amplitude increases.

The advantage is not that the random drive creates new thermal physics. The advantage is that it may excite many spatial and temporal modes in one measurement and allow them to be separated statistically through known input correlations.

17.2 Multiple-detector thermal measurements

Several response detectors may be placed at different spatial positions, scattering vectors, or probe geometries. Their signals can be written

$$\boldsymbol{Y}(t) = \begin{pmatrix} Y_1(t) \\ Y_2(t) \\ \vdots \\ Y_M(t) \end{pmatrix}. \tag{90}$$

The input–output correlation matrix is

$$\boldsymbol{C}_{uY}(\tau) = \begin{pmatrix} C_{uY_1}(\tau) \\ C_{uY_2}(\tau) \\ \vdots \\ C_{uY_M}(\tau) \end{pmatrix}. \tag{91}$$

The detector–detector matrix is

$$\boldsymbol{C}_{YY}(\tau) = [C_{Y_m Y_n}(\tau)]_{m,n}\,. \tag{92}$$

The asymmetry between positive and negative lag may reveal ordered response or propagation. The known drive is still needed to assign causal direction reliably.

18 Relation to fluctuation–dissipation ideas

The fluctuation–dissipation theorem relates equilibrium fluctuations to linear response under specific conditions. Schematically, a susceptibility or dissipative response can be related to an equilibrium correlation spectrum.

For a classical variable A near thermal equilibrium, one commonly encounters relations of the form

$$S_{AA}(\Omega) \propto \frac{k_{\mathrm{B}}T}{\Omega}\,\mathrm{Im}\,\chi_{AA}(\Omega), \tag{93}$$

with the exact prefactors and symmetrization depending on conventions and on the variables involved.

The proposed rotating-speckle experiment is not automatically an application of the fluctuation–dissipation theorem. The optical intensity fluctuations are externally generated, and a laser-heated sample is generally driven away from equilibrium.

The closer analogy is system identification with stochastic excitation:

$$\text{known random drive} + \text{drive–response cross-correlation} \longrightarrow \text{linear response function.} \tag{94}$$

This resembles the logic of fluctuation–dissipation methods because correlations reveal response functions, but the random drive is imposed rather than arising from spontaneous thermal equilibrium fluctuations.

The distinction should remain explicit:

- **Equilibrium FDT:** spontaneous equilibrium fluctuations determine linear response.

- **Stochastic system identification:** a measured external random drive is used to determine the causal response.

If the optical perturbation is weak enough for linear response and the unperturbed sample remains near equilibrium, the measured response function can still be compared with equilibrium transport theory. Strong driving, nonlinear response, or a nonequilibrium steady state requires a more general framework.

19 Proposed staged experimental program

19.1 Stage 1: scaling of single-detector autocorrelations

Measure $g^{(2)}(\tau)$ for several

$$q, \qquad \omega, \qquad R. \tag{95}$$

Test, in order,

$$G^{(2)} = F_{q,R}(\omega\tau), \tag{96}$$

$$G^{(2)} = F_q(R\omega\tau), \tag{97}$$

$$G^{(2)} = F\left(\frac{R\omega\tau}{\ell_c(q)}\right). \tag{98}$$

19.2 Stage 2: recurrence

Add an angular encoder or index pulse. Test

$$I(\theta + 2\pi, q) \approx I(\theta, q) \tag{99}$$

and measure recurrence over many revolutions.

19.3 Stage 3: two-detector correlations

Measure

$$C_{11}(\tau), \qquad C_{22}(\tau), \qquad C_{12}(\tau). \tag{100}$$

Determine whether the two autocorrelations obey the same scaled master law and how much independent information remains in the cross-spectrum and coherence.

19.4 Stage 4: reference-channel measurement

Split off a reference signal with a fiber coupler or beamsplitter. Measure

$$C_{\mathrm{ref},1}(\tau), \qquad C_{\mathrm{ref},2}(\tau). \tag{101}$$

Use these measurements to separate source fluctuations from diffuser and sample dynamics.

19.5 Stage 5: controlled input discrimination

Apply several known optical inputs $\boldsymbol{x}$ and test whether features derived from

$$I(\theta, q_m; \boldsymbol{x}) \tag{102}$$

improve discrimination or parameter estimation.
Compare

- raw angle-binned intensities,
- angular Fourier coefficients,
- autocorrelation features,
- auto- and cross-correlation features,
- reduced features selected by SVD or regularization.

19.6 Stage 6: driven thermal response

Use the measured or calibrated random optical field as a heating drive. Measure one or more thermal or structural responses and estimate transfer functions using input–output cross-correlations.

20 Primary initial predictions

The initial data-analysis code should treat the following as explicit testable predictions.

1. At fixed q and R,
$$\tau_c \propto \frac{1}{\omega}. \tag{103}$$
2. At fixed q and R, normalized curves collapse as
$$G^{(2)}(\tau, q; \omega, R) = F_{q,R}(\omega\tau). \tag{104}$$
3. If tangential displacement is the governing coordinate,
$$\tau_c \propto \frac{1}{R\omega} \tag{105}$$
and
$$G^{(2)} = F_q(R\omega\tau). \tag{106}$$
4. The empirical decorrelation length is
$$\ell_c(q) = R\omega\tau_c. \tag{107}$$
5. A possible global master law is
$$G^{(2)} = F\left(\frac{R\omega\tau}{\ell_c(q)}\right). \tag{108}$$

6. A possible, but not assumed, special case is

$$\ell_c(q) \propto q^{-1}, \tag{109}$$

leading to

$$G^{(2)} = F(qR\omega\tau). \tag{110}$$

7. The fitted decay exponent α should be approximately invariant under a true one-parameter collapse. A systematic dependence of α on ω, R, or q indicates that the curve shape is changing and that a simple rescaling is incomplete.

8. Collapse of autocorrelations does not imply that intensity traces at different q are identical. This must be tested with cross-correlation, coherence, covariance-rank, or mutual-information measurements.

21 References

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